

Markov Chain Monte Carlo

Handwritten Lecture Notes

ASM Course

Week-12, L01 MCMC

Topics: Chi-Squared Test · Transition Matrices · Stationary Distributions · Detailed Balance ·
Metropolis Hastings Algorithm

Markov Chain Monte Carlo

Problem Statement

Want to generate random variables from a distribution f .

$$X \sim f, \quad E[h(X)].$$

We want:

$$\frac{1}{n} \sum_{i=1}^n h(X_i), \quad X_1, \dots, X_n \sim f$$

However, we cannot generate directly from f (or from other distributions).

What you can do is find a Markov chain which has f as its stationary distribution.

The idea:

$$\frac{1}{n} \sum_{i=1}^n h(Y_i) \rightarrow E_f[h(X)]$$

Three main steps:

1. Example: χ^2 test for independence
2. How to build the chain
3. The convergence

Chi-Squared Test for Independence

Example: χ^2 test for independence (row and column attributes are independent).

$$E_{ij} \geq 5, \quad \sum_i (E_{ij} - D_{ij})^2 / E_{ij} \sim \chi_{(m-1)(k-1)}^2$$

Table representation:

$$\begin{array}{c|cc} & c_1 & c_k \\ \hline & & \end{array} \quad m \times k$$

You can generate all tables that have $r_1, \dots, r_m$ as row sums and $c_1, \dots, c_k$ as column sums. Then for each table you can find χ^2 .

If observed χ^2 is on the tail of this histogram then reject H_0 .

If there are M possible tables, we are putting prob p_n on each such table and then finding the $(1 - \alpha)$ -th quantile of the distribution (f).

Building the Chain

$T_0 \rightarrow T_1$ (all non zero elements): Description of T_1 : Pick 2 rows (i, i') and 2 columns $[j, j']$ at random.

$$t_{ij}^b \rightarrow z_{ij}^{-1}$$

$$z_{ij} \rightarrow z_{ij}^{-1}$$

$$x'_j \rightarrow x_j^{-1}$$

$$x_j^{-1} \rightarrow x_j^{+1}$$

$m \cdot (m-1) \cdot k \cdot (k-1)$ ways

Transition Matrices and Stationary Distributions

Markov Chain Fundamentals

Markov chain $P = ((P_{ij})) \rightarrow$ Transition matrix

Stationary distribution Π :

$$\Pi P = \Pi$$

$$\sum_i P_{ij} = \Pi_j, \quad \forall j$$

Problem: Given Π , how to find a P which is a transition matrix that satisfies $\Pi P = \Pi$.

The algorithm is **Metropolis Hastings**.

$$p(x, y) = \left(\frac{w(y)}{w(x)} \wedge 1 \right) / d$$

$$p_{ii} = 1 - \sum_{j \neq i} p_{ij}$$

To show **This is a transition matrix**.

Verification of Properties

(2) $\Pi P = \Pi$ is satisfied.

(i) Requires $\sum_j |p_{ij}| \leq 1$. This is true as $p_{ij} \leq 1/d \forall i \neq j$ and there are $d-1$ of them.

Detailed Balance Equation

Detailed balance equation:

$$\Pi(i)p_{ij} = \Pi(j)p_{ji}, \quad \forall i, j$$

(3) We will show that P satisfies detailed balance equation.

(4) If P satisfies detailed balance equation, then P satisfies $\Pi P = \Pi$.

$$\begin{aligned} P(i, j) &= \Pi(j)P_{ij} \sum \Pi_j P_{ij} \\ &= \Pi_j \sum b_{ii} \quad (\text{banach matrix}). \end{aligned}$$

Pf 3

$$\begin{aligned} \Pi_i p_{ij} &= \Pi_i \left(\frac{\Pi_j}{\Pi_i} \wedge 1 \right) / d = (\Pi_j \wedge \Pi_i) / d \\ \Pi_j p_{ji} &= \Pi_j \left(\frac{\Pi_i}{\Pi_j} \wedge 1 \right) / d = (\Pi_i \wedge \Pi_j) / d \end{aligned}$$

Pf (4).

$$\begin{aligned} \sum_i \pi_i P_{ij} &= \sum_i \pi_i p_{ij} \\ &= \pi_j \sum_i p_{ji} \\ &= \pi_j \cdot 1 \quad (\text{by normalization condition}). \end{aligned}$$

Given π , we have exhibited a P that is a transition matrix and has π as its stationary distribution.

If $\tilde{\pi}_i = c w_i$ and c is unknown, even then the Metropolis–Hastings algorithm works.

In the definition of p_{ij} , we need

$$\frac{\tilde{\pi}_j}{\tilde{\pi}_i} = \frac{c w_j}{c w_i} = \frac{w_j}{w_i}.$$

So we do not need c . It is enough to know w 's.

In our example, π is uniform. So the ratios satisfy

$$\frac{w_j}{w_i} = 1 \quad \forall i, j.$$

But this is still not enough, as we do not know d .